

```
1  library ieee;
2  use ieee.std_logic_1164.all;
3
4  entity display_unit is
5      port( r_in:      in std_logic_vector(3 downto 0);
6            segments: out std_logic_vector(6 downto 0));
7  end;
8
9  architecture synth of display_unit is
10 begin
11     process (r_in) begin
12         case r_in is
13             when X"0" => segments <= "1111110";
14             when X"1" => segments <= "0110000";
15             when X"2" => segments <= "1101101";
16             when X"3" => segments <= "1111001";
17             when X"4" => segments <= "0110011";
18             when X"5" => segments <= "1011011";
19             when X"6" => segments <= "1011111";
20             when X"7" => segments <= "1110000";
21             when X"8" => segments <= "1111111";
22             when X"9" => segments <= "1111011";
23             when others => segments <= "0000000";
24         end case;
25     end process;
26 end;
```